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Sexth

2025

(Session : 2024-27)

Time : 3 hours

Full Marks : 100

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Answer from all the Groups as directed.

Group – A

(Compulsory)

1. Select the correct answer of the following multiple-choice questions : 2×10 = 20

(a) A spanning tree of a graph :

(i) Contains all edges

(ii) Contains all vertices and minimum possible edges

(iii) Contains cycles

(iv) Must be a complete graph

(b) In a complete graph K_n, the number of edges is :

(i) n^2

(ii) $n(n-1)/2$

(iii) $2n$

(iv) $n-1$

(c) A relation R on a set A is transitive if :

(i) $(a, b) \in R \Rightarrow (b, a) \in R$

(ii) $(a, b) \in R$ and $(b, c) \in R \Rightarrow (a, c) \in R$

(iii) $a = b$ for all pairs

(iv) None of these

(d) A relation R on a set A is symmetric if :

(i) $(a, b) \in R \Rightarrow (b, a) \notin R$

(ii) $(a, b) \in R \Rightarrow (b, a) \in R$

(iii) $(a, a) \in R$ for all $a \in A$

(iv) $(a, b) \in R \Rightarrow (a, a) \in R$

(e) Formula for permutations of n objects taken r at a time :

(i) $n! / (n-r)!$

(ii) $(n-r)! / n!$

(iii) n^r

(iv) $r!$

(f) The formula for combinations of n objects taken r at a time is :

(i) $n! / (n-r)!$

(ii) $n! / r! (n-r)!$

(iii) n^r

(iv) r^n

(g) The statement $(p \vee q) \wedge \neg q$ is true when :

(i) q is true

(ii) p is true and q is false

(iii) both p and q are true

(iv) p and q both false

(h) The implication $p \rightarrow q$ is false only when :

(i) p is true and q is false

(ii) p and q are true

(iii) p and q are false

(iv) p is false and q is true

(i) Two sets are said to be equal if :

(i) They have the same number of elements

(ii) They contain exactly the same elements

(iii) They are subsets of each other

(iv) Order is same

(j) If two sets have no common elements, they are :

(i) Equal

(ii) Subsets

(iii) Disjoint

(iv) Infinite

Group – B

Answer any **four** questions of the following : 5×4 = 20

2. Differentiate between directed and undirected graph.

3. How many ways can you form a committee of 3 people from 9 men and 7 women, where the committee must have 2 men and 1 women ?

4. How many distinct ways can the letters of the word "BANANA" be arranged ?

5. In a group of 50 people, 35 speak Hindi, 25 speak both English and Hindi and all the people speak at least one of the two languages. How many people speak only English and not Hindi ? How many people speak English ?

6. What is reflexive and irreflexive relation ? Give one example of each relation.

7. Use the Master theorem. Solve the recurrence relation.

$T(n) = 3T(n/2) + n^2$

8. Solve the recurrence relation Fn = 5Fn – 1 – 6Fn – 2, where F0 = 1 and F1 = 4.

Group – C

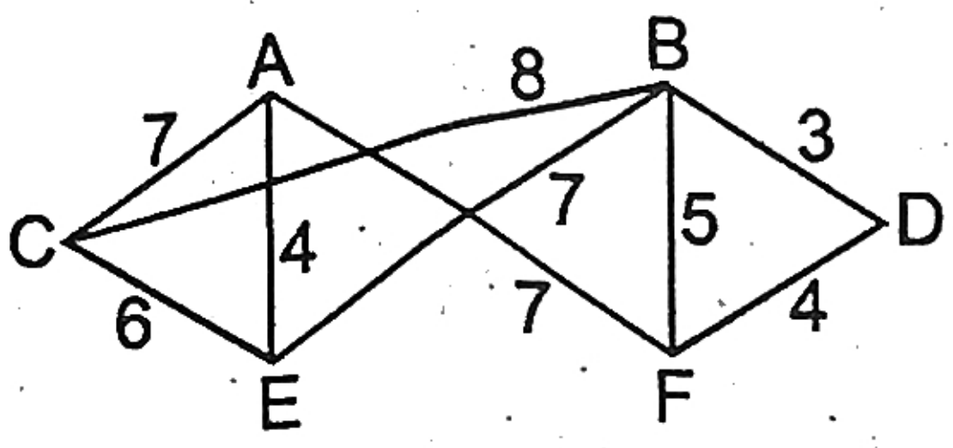
Answer any **two** questions of the following : 15×2 = 30

9. Solve the following recurrence relation using back substitution method :

$T(n) = 2T(n/2) + n, T(1) = 1$

10. What is function ? Explain the types of function with suitable example.

11. Find the minimum cost spanning tree of the graph given below using prim's algorithm :



12. Check whether the given propositional logic are tautology or not :

(a) $((a \rightarrow b) \wedge (b \rightarrow c)) \rightarrow (a \rightarrow c)$

(b) $(a \leftrightarrow b) \rightarrow (\sim b \rightarrow (a \wedge c))$